

1.6 RF EXCITATION & FLIP ANGLE

Using RF energy to tip the net magnetization away from B_0

When an RF pulse is applied at the Larmor frequency (f_0), the net magnetization (M_0) absorbs energy and tips away from the B_0 axis into the transverse plane.

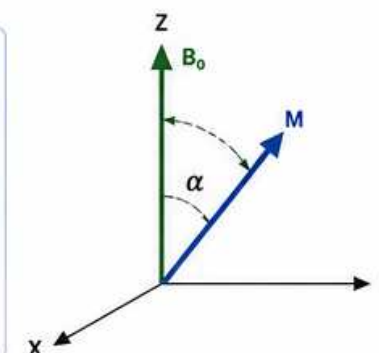
The amount of tipping is the **FLIP ANGLE** (α).

THE CONCEPT

- RF magnetic field (B_1) rotates in the transverse plane at the Larmor frequency.
- When $f_{RF} = f_0$ (resonance), the proton spins lock with B_1 and energy is absorbed.
- M_0 tips away from the +Z (B_0) axis by an angle α .
- α depends on B_1 amplitude and pulse duration (τ).

Flip angle equation:

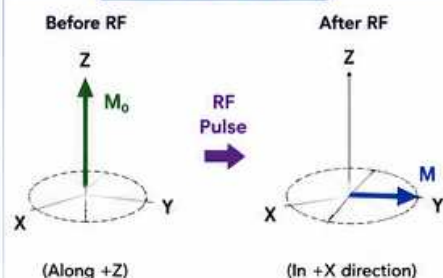
$$\alpha = \gamma B_1 \tau \quad (\text{radians})$$



B_0 = Main magnetic field (along +Z)
 B_1 = RF magnetic field (transverse)
 M_0 = Net magnetization at equilibrium (along +Z)
 M = Net magnetization after RF pulse (tipped by α)
 α = Flip angle
 τ = RF pulse duration
 γ = Gyromagnetic ratio

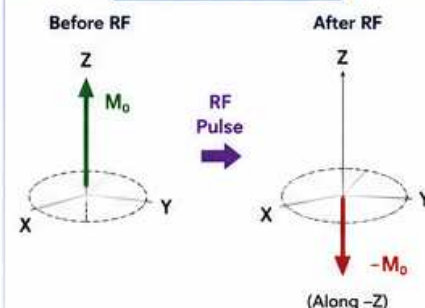
TIPPING THE NET MAGNETIZATION

$\alpha = 90^\circ$ PULSE



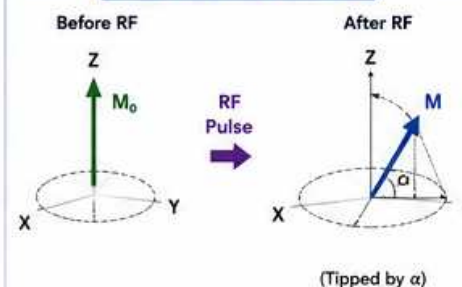
90° pulse tips M_0 into the transverse (XY) plane. Maximum transverse magnetization (M_{xy}).

$\alpha = 180^\circ$ PULSE



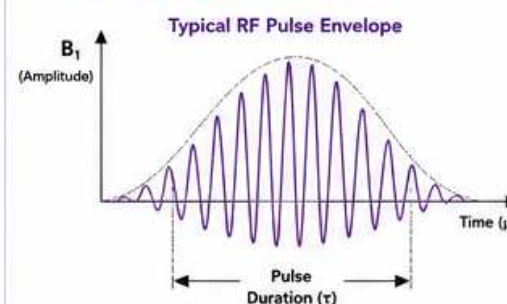
180° pulse inverts M_0 . No transverse magnetization.

$\alpha = \text{OTHER FLIP ANGLES}$



Any flip angle (α) between 0° and 180° produces a component in the XY plane.

RF PULSE: TIME DOMAIN



Key Points

- The area under the B_1 envelope ($B_1 \times \tau$) determines the flip angle.
- Longer pulse or higher B_1 → larger flip angle.
- Shorter pulse or lower B_1 → smaller flip angle.

Common Flip Angles

30°

(Partial)

60°

(Preparation)

90°

(Excitation)

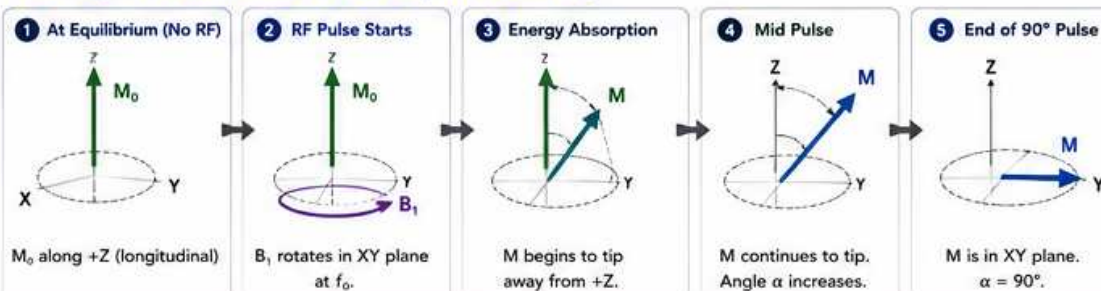
120°

(Inversion Recovery)

180°

(Inversion)

HOW M_0 TIPS DURING AN RF PULSE (90° EXAMPLE)



Remove RF → M precesses in the XY plane and induces signal in the receive coil.

ENGINEERING INSIGHT

- Flip angle accuracy is critical for image contrast.
- Transmit chain: Amplifier → Coils → Patient.
- B_1 field should be uniform for consistent flip angle everywhere.
- Flip angle errors lead to contrast loss and quantitative measurement errors.

FIELD SERVICE TIP

- Verify RF output using service mode or a phantom (e.g., 90° calibration).
- Check for B_1 inhomogeneity: Look for shading in flip angle maps or uniform phantom scans.
- Common causes: Faulty amplifier, detuned coils, poor cable connections, or patient loading.

COMMON MISCONCEPTION

"A higher RF power always means a 90° pulse."

Reality: Flip angle depends on B_1 amplitude AND pulse duration (τ). Too much power or wrong τ → wrong flip angle.

REAL-WORLD ANALOGY

Think of a spotlight:

- The brightness (B_1 strength) and the time it shines (τ) determine how far the light (M) moves from pointing straight up.



KEY TAKEAWAY

RF excitation tips the net magnetization by a flip angle (α). α is controlled by the strength and duration of the RF pulse at the Larmor frequency. This creates transverse magnetization, which is the source of the MRI signal.

WHAT'S NEXT? >>

Next: Section 1.7 – Resonance Condition

We'll see why matching RF frequency to the Larmor frequency is essential for efficient energy transfer.

